

CS & IT ENGINEERING

DISCRETE MATHS
COMBINATORICS



Lecture No. 05



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A whiteboard with a blue border and an orange base, displaying the topics to be covered.

TOPICS TO BE COVERED



01 Principle of derangement



02 Formula for derangement



03 Problems in derangement



COMBINATORICS



Arrangement:
Set: $\{1, 2, 3\}$

$3!$ {

1	2	3
1	3	2
2	1	3
2	3	1
3	1	2
3	2	1

COMBINATORICS



Derangement:

$$D_3 = 2$$

$S_1 \downarrow 2$ $S_2 \downarrow 3$ $S_3 \downarrow 1$
 $S_1 \downarrow 3$ $S_2 \downarrow 1$ $S_3 \downarrow 2$

S_1	S_2	S_3	
1	2	3	
1	3	2	X
2	1	3	X
2	3	1	✓
3	1	2	✓
3	2	1	X

no. of ways such that students are not seating @ right place.

Ans : 2 ways.

$\begin{cases} 2 & 3 & 1 \\ 3 & 1 & 2 \end{cases}$

COMBINATORICS



Derangement (D_n)

nothing is at right position.

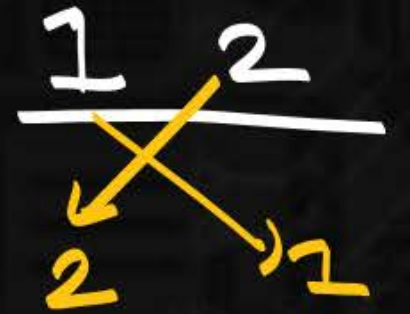
$$D_1 = 0$$

$$D_2 = 1$$

$$D_3 = 2$$

$$D_4 =$$

$$D_5 =$$



COMBINATORICS



$$D_3 = 3! - \text{at least 1 element @ right position.}$$

$$D_4 = 4! - \text{at least 1 element @ right position.}$$

$$D_4 = 4! - | A_1 \cup A_2 \cup A_3 \cup A_4 |$$

COMBINATORICS



1 2 3 4.

$A_1 = \text{no. of ways where 1 element @ right}$

$A_2 = \text{no. of ways} \text{ --- } 2 \text{ --- } "$

$A_3 = \text{no} \text{ --- } 3 \text{ --- } "$

$A_4 = \text{--- } 4 \text{ --- } "$

COMBINATORICS



$$A_1 \cup A_2 \cup A_3 \cup A_4 = \frac{3! \quad 3! \quad 3! \quad 3!}{A_1 + A_2 + A_3 + A_4}$$

$$= \sum A_i \cap A_j + \sum A_i \cap A_j \cap A_k - A_1 \cap A_2 \cap A_3 \cap A_4$$

$$= \left(\underline{4 \cdot 3! - 4c_2 \cdot 2!} + 4c_3 \cdot 1! - 1 \right)$$

COMBINATORICS



$$A_1 = 3! \quad A_3 = 3!$$

$$A_2 = 3! \quad A_4 = 3!$$

$$A_1 \cap A_2 = 2!$$

$$A_2 \cap A_3 = 2!$$

$$\sum A_i \cap A_j = {}^4C_2 \times 2!$$

$$\begin{array}{c} \underline{1 \ 2 \ 3} \\ 1 \ 2 \ 3 \\ 1 \ 2 \ 3 \\ 1 \ 2 \ 3 \end{array} \left\{ \begin{array}{c} 4 \\ 4 \\ 4 \\ 4 \end{array} \right.$$

11.

$$A_1 \cap A_2 \cap A_3$$

$$\sum A_i \cap A_j \cap A_k = {}^4C_3 \cdot 1!$$

$$\underline{A_1 \cap A_2 \cap A_3 \cap A_4 = 1}$$

all element @ right
position = 1 way.

$$\boxed{1 \ 2 \ 3 \ 4}$$

COMBINATORICS



$$D_4 = 4! - |A_1 \cup A_2 \cup A_3 \cup A_4|$$

$$D_4 = 4! - [4 \cdot 3! - 4C_2 \cdot 2! + 4C_3 \cdot 1! - 1]$$

$$(4 \cdot 3! = 4!)$$

$$= \cancel{4!} - \cancel{4 \times 3!} + \underline{4C_2 \times 2!} - 4C_3 \times 1! + 1 = 4C_2 \cdot 2! - 4C_3 \cdot 1! + 1$$

$$\left(1 = \frac{4!}{4!}\right)$$

$$= \frac{4!}{\cancel{2! \times 2!}} \times \cancel{2!} - \frac{4!}{\cancel{3!} \cdot \cancel{1!}} \times \cancel{1!} + \frac{4!}{4!} = 4! \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right]$$

COMBINATORICS



$$D_4 = 4! \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right] = 9$$

$$D_n = n! \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \cdots \frac{(-1)^n}{n!} \right]$$

$$D_1 = 0$$

$$D_2 = 1$$

$$D_3 = 2$$

$$D_4 = 9$$

$$D_5 = 44$$

$$D_6 = 265$$

COMBINATORICS



GATE

$B_1 \quad B_2 \quad B_3 \quad B_4 \quad B_5$
 $C_1 \quad C_2 \quad C_3 \quad C_4 \quad C_5$

$B_i \rightarrow C_i$

$$D_5 = 44$$

How many 1-1 functions are possible set of 6 elements so that no element is mapped to itself?

$$D_6 = 265$$

COMBINATORICS



How many 1:1 functions are possible with 6 elements such that no element is mapping to itself?

$$D_6 = 265$$

How many 1:1 functions are possible with 6 elements such that at least 1 element is mapped to itself?

$$D_6 = 6! - \text{at least 1 mapped to itself}$$

$$\text{at least 1 element mapped to itself} = 6! - D_6$$

COMBINATORICS



5 students $S_1 S_2 S_3 S_4 S_5$
 5 places $P_1 P_2 P_3 P_4 P_5$

$${}^5C_2 \times D_3 = 10 \times 2 = 20$$

3. exactly 2 students are correctly seating.



1) no student is correctly seating.

2) at least 1 student is correctly seating.

4. at most 1 student is correctly seating.

$$D_5 + {}^5C_1 \cdot D_4 = 44 + 5 \times 9 = 89$$

COMBINATORICS



at most 1 student is correctly placed.

Exactly 0 student is correctly placed. or Exactly 1 student is correctly placed.

$$D_5 + {}^5C_1 \cdot D_4.$$

$$\begin{aligned} &44 + 5 \times 9 \\ &= 44 + 45 = 89. \end{aligned}$$

COMBINATORICS



$$e^x = 1 + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} \dots$$

$$x = -1.$$

$$e^{-1} = 1 + \frac{(-1)^1}{1!} + \frac{(-1)^2}{2!} + \frac{(-1)^3}{3!} + \frac{(-1)^4}{4!} \dots$$

$$e^{-1} = \cancel{1} - \cancel{1} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \dots$$

$$e^{-1} = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{(-1)^n}{n!}$$

COMBINATORICS



$$D_n = n! \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + \frac{(-1)^n}{n!} \right]$$

$$D_n \approx n! \times e^{-1} \quad (n > 6)$$

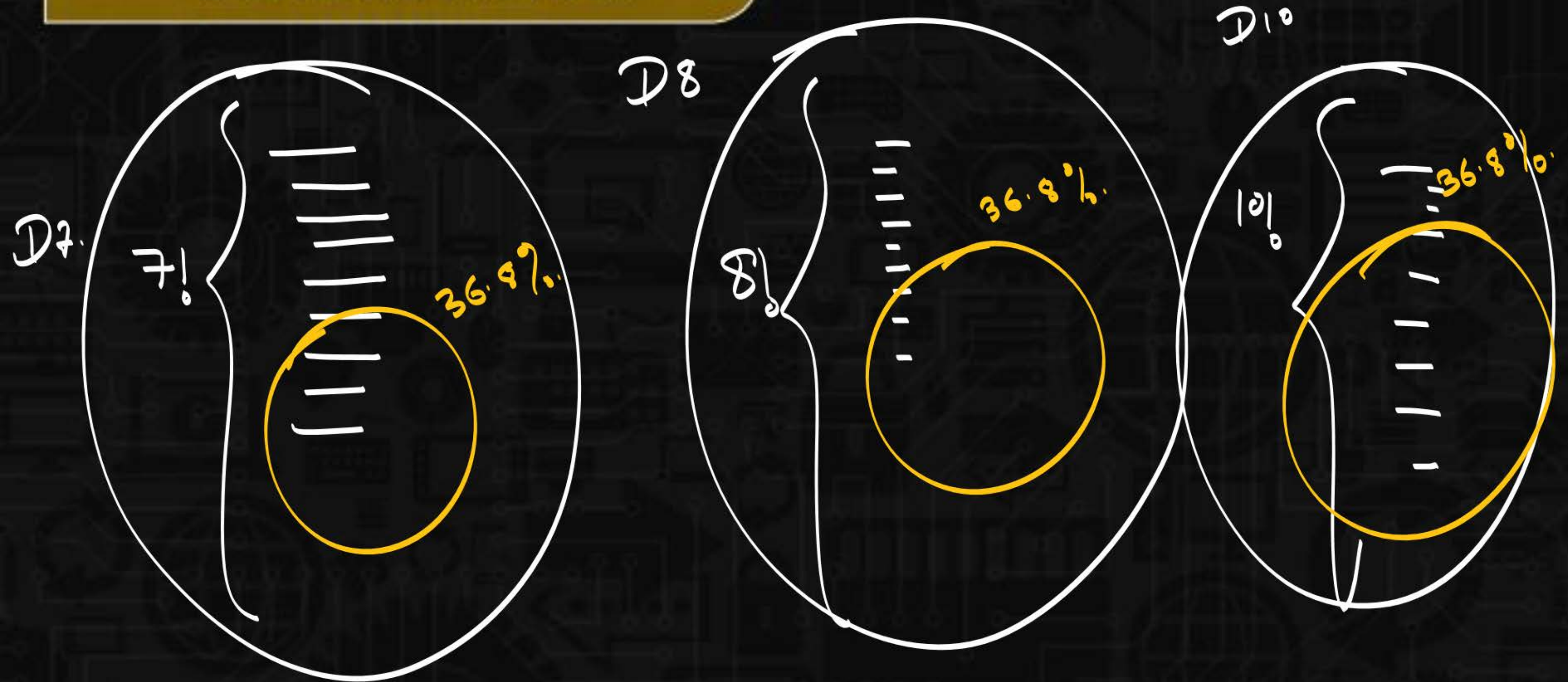
$$D_n \approx n! \times 0.368$$

$$e^{-1} = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + \frac{(-1)^n}{n!}$$

$$e = 2.718$$

$$1/e = e^{-1} = 0.368$$

COMBINATORICS



COMBINATORICS



$$Dn \equiv n! \times 0.368 \quad \frac{Dn}{n!} \approx 0.368$$

$$D7 \equiv 7! \times 0.368$$

$$\frac{D7}{7!} = 0.368$$

$$\frac{D8}{8!} = 0.368$$

